MILESTONE-2 REPORT

CliniScan - Lung-Abnormality Detection on Chest X-rays using AI

Model Training and Evaluation

Medical Image Analysis Pipeline for Chest X-ray Abnormalities Detection

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1. Introduction

Medical image analysis plays a critical role in assisting radiologists in detecting diseases at an early stage. Chest X-rays are widely used for diagnosing lung abnormalities such as pneumonia, fibrosis, nodules, and lung opacity. However, analyzing large volumes of X-ray images manually can be time-consuming and prone to human error.

> *The objective of this milestone is to develop a machine learning pipeline capable of detecting abnormalities in chest X-ray images.*

The system includes two major components:

- **Classification Model:** Predicts which diseases are present in the image.
- **Object Detection Model:** Identifies the location of abnormalities using bounding boxes.

The dataset used in this project is the VinBigData Chest X-ray Abnormalities Detection dataset. The images are processed from DICOM to PNG format. The classification model uses ResNet50, while the object detection model leverages YOLOv8 for efficient and accurate localization.

2. Dataset Description

The dataset used in this project is the VinBigData Chest X-ray Abnormalities Detection dataset, containing annotated chest X-ray images used for training machine learning models.

Feature	Description
Dataset Name	VinBigData Chest X-ray Dataset
Image Format	DICOM
Converted Format	PNG
Total Images	~15,000
Annotation Type	Bounding Boxes
Number of Classes	Multiple lung abnormalities

Sample Abnormalities

- Lung Opacity
- Consolidation
- Fibrosis
- Mass
- Nodules
- Pleural Effusion
- Pneumothorax

3. Data Preprocessing

The dataset initially consists of DICOM medical image files, which must be converted into a format suitable for deep learning models.

Preprocessing Steps

- **DICOM to PNG Conversion:** Pixel data extracted, intensity values normalized, and saved as PNG.
- **Normalization:** Pixel values scaled to a range of 0–255.
- **Cropping:** Removal of unnecessary black borders around X-ray images.
- **Image Resizing:** Classification (224 × 224), Detection (640 × 640).
- **Contrast Enhancement:** Applied CLAHE (Contrast Limited Adaptive Histogram Equalization).
- **Noise Reduction:** Applied Gaussian Blur to remove minor noise.

4. Dataset Structure

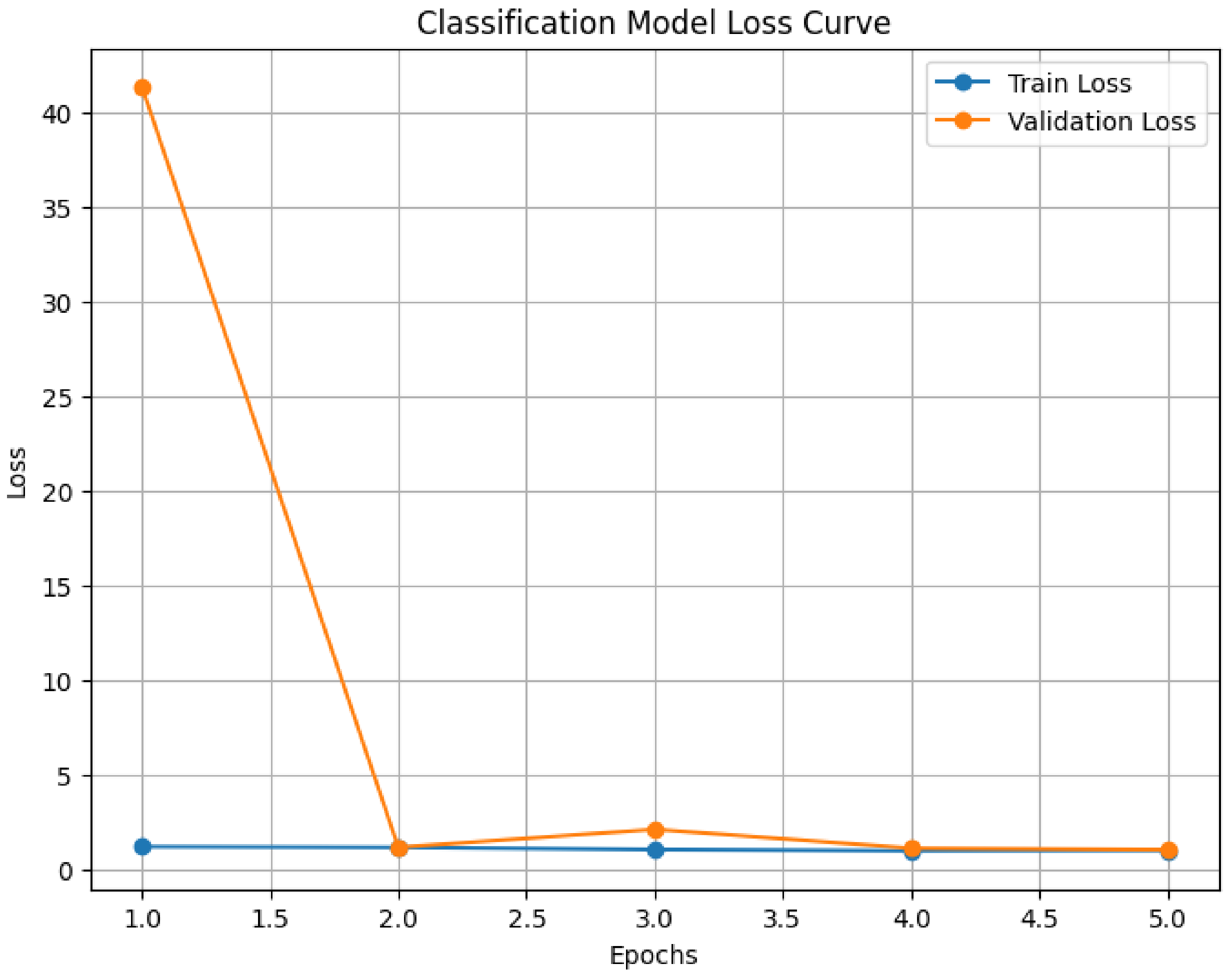
After preprocessing, the dataset was organized in the following hierarchy to support standard ML training frameworks:

```
dataset/
 ├── images/
 │   ├── train/
 │   ├── val/
 │   └── test/
 └── labels/
     ├── train/
     ├── val/
     └── test/
```

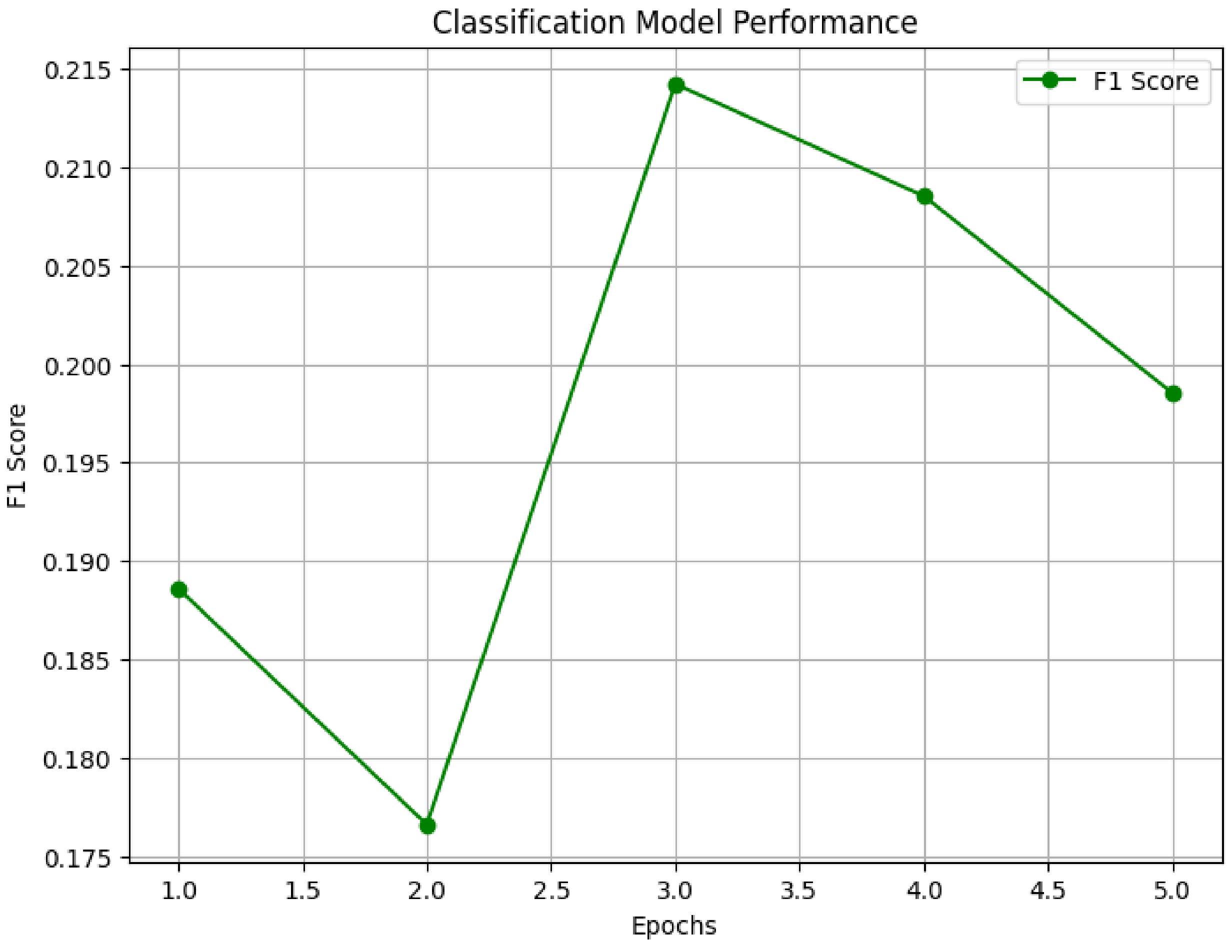
Example: image_001.png paired with image_001.txt.

5. Classification Model (ResNet50)

The classification stage identifies which abnormalities exist. ResNet50 was chosen due to its high performance and residual connections that prevent vanishing gradient problems. The final layer was modified for multi-label classification.



5. Classification Model (ResNet50)



6. Classification Training Setup

Parameter	Value
Model	ResNet50
Image Size	224 × 224
Batch Size	32
Optimizer	Adam
Learning Rate	0.001
Loss Function	BCEWithLogitsLoss
Epochs	5
Hardware	GPU (Tesla T4)

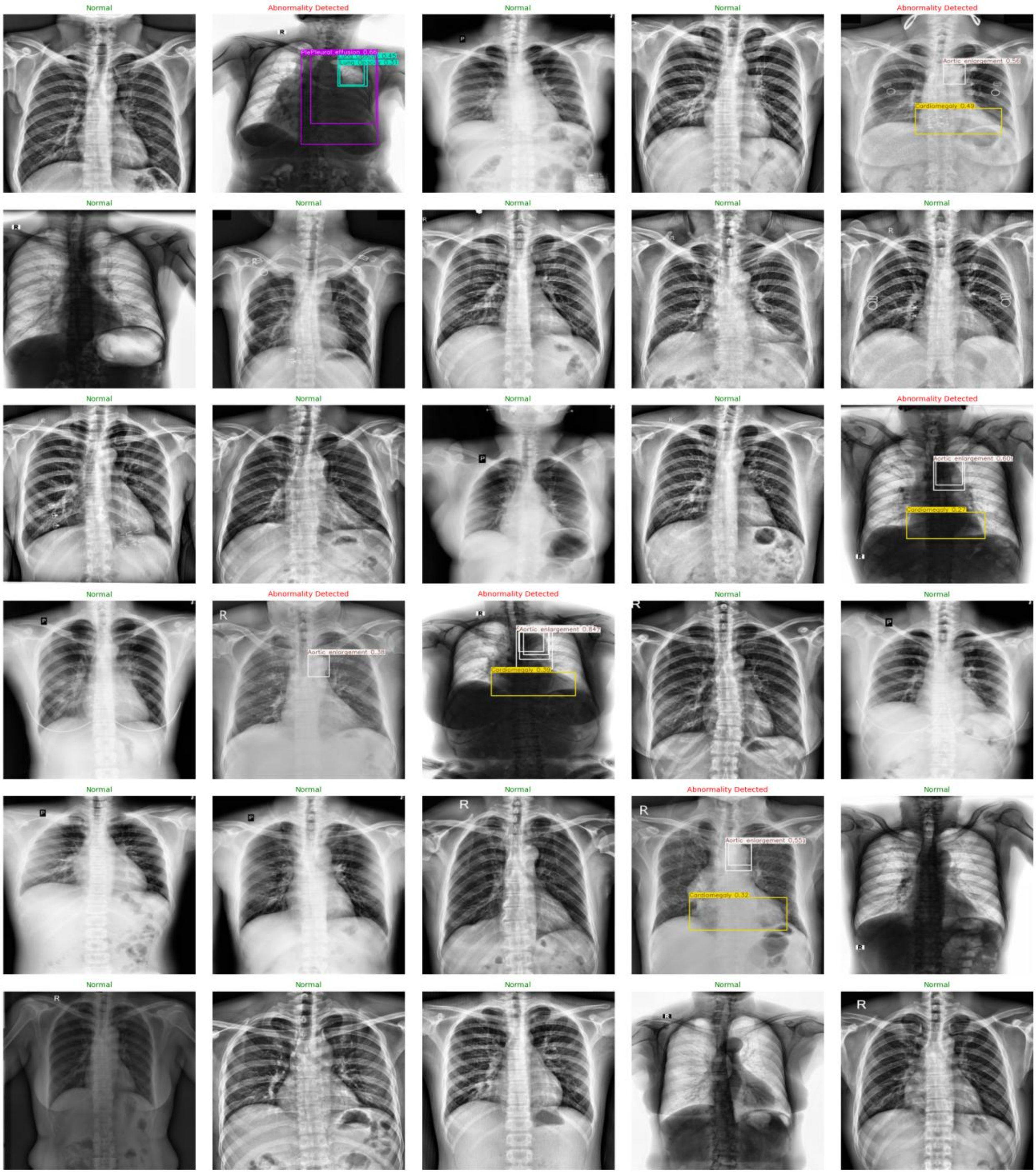
7. Classification Results

Metric	Value
Train Loss	**~0.82**
Validation Loss	**~0.88**
F1 Score	**~0.28**

The best performing model was saved as `best_classification_model.pth`.

8. Object Detection Model (YOLOv8)

YOLOv8 (You Only Look Once) was implemented for high-speed, real-time object localization. It predicts bounding boxes, class labels, and confidence scores simultaneously.



9. Annotation Format

Annotations follow the standard YOLO format: `class_id center_x center_y width height`. All values are normalized (0 to 1).

5 0.2571614583333333 0.35260416666666666 0.041232638888888889 0.03229166666666667
5 0.25390625 0.3541666666666667 0.033854166666666664 0.027777777777777776
5 0.2575954861111111 0.3545138888888889 0.03342013888888889 0.034027777777777775
12 0.2575954861111111 0.3545138888888889 0.03342013888888889 0.034027777777777775

10. Detection Training Setup

Parameter	Value
Model	YOLOv8m
Image Size	640
Epochs	20
Batch Size	8
Framework	Ultralytics YOLO

11. Evaluation Metrics

- **Precision:** Accuracy of predicted boxes.
- **Recall:** Percentage of true abnormalities captured.
- **mAP@50:** Mean Average Precision at IoU 0.5.
- **mAP@50-95:** COCO-standard metric.

12. Detection Results

Metric	Value
Precision	**0.43**
Recall	**0.24**
mAP@50	**0.22**
mAP@50-95	**0.10**

13. Hybrid Pipeline

The system integrates both models in a sequential workflow:

1. **Input:** Chest X-ray image is received.
2. **Step 1:** Classification model predicts disease presence.
3. **Step 2:** If disease detected (True), the Detection model runs.
4. **Output:** Localized bounding boxes and labels are returned.

14. Challenges Faced

- Handling of complex DICOM formats and conversion overhead.
- Significant class imbalance within the medical dataset.
- Low recall observed for rare lung abnormalities.
- Substantial computational time required for training deep models.

15. Future Improvements

- Extend detection training to 50–80 epochs for convergence.
- Experiment with larger architectures (YOLOv8l or YOLOv8x).
- Implemented weighted loss functions for better class balancing.
- Ensemble methods combining multiple model predictions.

16. Deliverables

Deliverable	Description
Trained Classification Model	best_classification_model.pth
Trained Detection Model	best.pt
Preprocessed Dataset	PNG conversions
Visualizations	Bounding box sample outputs

17. Conclusion

In this milestone, a complete deep learning pipeline was developed for detecting abnormalities in chest X-ray images. The results demonstrate the potential of deep learning models to assist radiologists in medical image analysis. Future iterations focusing on architecture depth and data balancing will further enhance system accuracy.